Appendix H
Physics for the Body: Measurements, Imaging, and Treatment

This appendix addresses physics on the body. It begins with a short overview of the physics of measurements and medical treatments that use light, ultrasound, x-rays, radioactivity, and magnetic resonance imaging. It concludes with locators for topics in the physics of measurements and of treatment that are presented in the main text. There are relatively few such examples because this is not the emphasis of this book.

This area of physics devoted to diagnosing and treating disease is known in the United States as "medical physics" [2]. It is subdivided by some into diagnostic and therapeutic radiology, diagnostic and therapeutic nuclear medicine, ultrasound, magnetic resonance imaging, and the applications of lasers for diagnosis and treatment [2], and by others in a slightly different manner, into the physics of radiation therapy, the physics of imaging, the physics of nuclear medicine, and health physics [4]. Medical physics is addressed in [3] and many resources in this area are listed in [1, 2]. Resources for the more narrow area of radiation therapy can be found in [4].

H.1 Overview

Visible light is widely used for medical inspection and visible and ultraviolet light are widely used for medical treatment. Fiber optics, in which light is guided through flexible fibers, are used in endoscopes to deliver light to internal regions and to collect scattered light to form an image on a charge-coupled device (CCD) array in the endoscope tip. These are used for inspection diagnostics and also for inspection during surgery or a procedure. The rigid sigmoidoscope can inspect only the lower region of the colon and the flexible colonoscope can also inspect the upper regions of the colon; both also have forceps or snares to remove polyps for biopsy. Laparoscopes are used for several types of surgery and procedures, including stomach surgery, hernia repair, gallbladder removal, and gynecological procedures. Arthroscopes are used for surgery on articular joints, such as knees, ankles, and shoulders. Other endoscopes include: bronchoscopes (for the bronchial passage in the lung),

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cystoscopes (urinary bladder), thoracoscopes (chest cavity), laryngoscopes (larynx), and angioscopes (blood vessels).

Lasers with a wide range of wavelengths, powers and, for pulsed lasers, pulse energies are used for many types of surgical treatments to make incisions and remove tumors and other tissues without bleeding and for sculpting, and are expected to be utilized increasingly for other treatments. They are usually employed for heating, with gentle heating, photocoagulation, and photovaporization at increasingly high laser intensities; however, some applications use photon absorption to break bonds without much heating, as in laser ablation and photo-treatments. Lasers are widely used in ophthalmology, such as for cornea sculpturing (PRK and LASIK), addressing retina diseases (e.g. to coagulate thin-walled, proliferating capillaries that easily rupture, which is common in diabetics, and to treat and repair holes and tears that can lead to progressive damage), treating glaucoma (to create holes to drain the aqueous humor to relieve the excess humor pressure), and treating cataracts (by cleaning opaque regions that are sometimes present after lens replacement). Lasers are also routinely used in gynecological procedures and dermatology, as in removing port-wine stains birthmarks and tattoos. In recent photodynamic treatments tumors are selectively killed by toxic species. These toxins are released locally by laser irradiation which dissociates photosensitizers that are introduced into the body and collect in tumor cells.

Ultrasound is widely used for imaging, as of fetuses. It is used in echocardiography to image the heart and major vessels and to analyze blood flow by the Doppler shifting of the frequency of sound scattered from red blood cells. Ultrasound is also used to break up kidney stones.

The transmission of x-rays by the body provides two-dimensional images (by the integrated absorption of x-rays along linear paths in the body) to obtain images not possible by visible light or ultrasound. Bones are often highlighted in x-rays because they absorb x-rays more strongly than do soft tissues. Specific tissues can be highlighted after the injection of and subsequent selective and transient incorporation of nontoxic, radiopaque substances. Such treatment utilizes high-Z atoms, such as barium or iodine, which absorb strongly at the x-ray energies commonly used. In computed tomography (CT), transmission is measured at many different angles, to arrive at a three-dimensional reconstruction by using tomographic analysis procedures.

Radioactivity is used both for imaging and treatment. In radionuclide imaging, radioactive materials are introduced locally in the body and some of their decay products, usually gamma rays, are detected and imaged outside the body. Gamma cameras are used for two-dimensional imaging of gamma rays, while tomographic methods are used to obtain three-dimensional reconstructions, as in single photon emission computed tomography (SPECT) and positron emission tomography (PET). Gamma rays that are emitted by a specific radionuclide are detected in SPECT, while in PET the 511 keV photons produced when positrons emitted by a specific radionuclide combine with nearby (within $\sim$1–2 mm) electrons are detected. Compared to x-ray imaging, radionuclide imaging has poorer resolution, but excellent contrast and can be used to analyze body function and not to just provide anatomical information.

In radiation treatment, tumors are killed by ionizing radiation from one of several sources. This includes beam sources such as linear accelerators or linacs, which produce high-energy x-rays and electrons, and (the relatively few) particle accelerators that produce neutrons. External sources include the gamma-ray emitting cobalt-60 (which, compared to linacs, have lower beam quality, limited duration, and only one wavelength), brachytherapy sources (sealed radioactive sources placed very near the tumor), and unsealed sources (radionuclide sources that are injected or swallowed).

Magnetic resonance imaging (MRI) uses the principle of nuclear magnetic resonance (NMR) in which an applied magnetic field causes the energy levels corresponding to the spin of a nucleus to split into different energies and electromagnetic radiation (resonantly tuned to a frequency corresponding to the separation of these levels) is applied and is absorbed and then re-emitted. Using a method known as the spin echo, two such electromagnetic pulses are incident on the body and a third one is emitted and then detected. The intensity of the emitted radiation depends on how fast the local medium relaxes the excited spins from one level to another; this is characterized by relaxation times called T1 and T2; the shorter these times the smaller the signal. MRI usually probes the spin states of protons (hydrogen nuclei). One key to MRI is that the values of T1 and T2 determined from this analysis characterize the concentrations and chemical environments of specific nuclei in local molecules and materials. This distinguishes one environment from another and provides the contrast in the image. The second key feature is that this information can be obtained in a spatially resolved manner. This spatial imaging occurs because the resonant frequency of the absorbed (and then emitted) radiation is determined by the local magnetic field, and the applied field has a spatial gradient (on a top of a background value) that is swept in time, and so the location of the resonance moves and can be tracked. In MRI the spatial resolution ranges from 0.25 to several mm. The electromagnetic pulses used in MRI oscillate at frequencies in the radio frequency (RF) region. The interaction with the nuclear spins is through the oscillating magnetic field in the electromagnetic pulse. In contrast, in all of the interactions of electromagnetic radiation with matter with visible light (above and elsewhere in the book, such as light absorption in the retina rods and cones) and x-rays (above) the interaction with matter is through the oscillating electric field of the electromagnetic field. MRI is less invasive than methods employing x-rays and radionuclides.

H.2 Locator

We now note the several measurements of the body described in the text, along with the fewer text references to treating the body. This encompasses a very small fraction of the medical physics topics we just overviewed; it also includes properties measurements and medical examination techniques that were not covered in the appendix overview.

Several problems in Chap. 1 describe how to measure macroscopic properties of the body. Problem 1.39 and Fig. 1.18 show how to measure the volume of body

segments. Determining body fat content is described in Problem 1.43 and Fig. 1.19. Problems 1.45 and 1.46 and Fig. 1.20 show how to determine the body center of mass and the mass of distal segments.

In Chap. 6, ways to measure metabolic rates are depicted (in Figs. 6.6 and 6.7). The protocols used to assess cardiovascular and cardiorespiratory capacity are also described (Table 6.24).

Measurements concerning the heart and blood flow performance, the focus of Chap. 8, are presented in several chapters. Chapter 7 describes how to measure blood pressure with a sphygmomanometer (Fig. 7.2) and the use of palpation to estimate the internal pressuse of a blood vessel (Problem 7.14, Fig. 7.25). This chapter and the next also present the physics of intravenous injection (with a hypodermic syringe, Problem 7.20, Fig. 7.26) and infusion (Problem 8.6, Fig. 8.64). Irregularities in cardiac valve openings and closings and in flow are presented in Chap. 8 (Fig. 8.5) and in Chap. 10 (Fig. 10.55, for heart disorders), which also has problems detailing how echocardiography uses ultrasound to diagnose heart disorders and blood flows (Problems 10.22 and 10.23). Problem 8.51 outlines the use of ballistocardiography to measure the ejection of blood from the heart. Use of EKGs to monitor the heart is described in Chap. 12 (see below).

Chapter 9 describes how to use a spirometer to measure lung volumes during different stages of breathing (Fig. 9.13).

Chapter 10 explains how listening to sound, called auscultation, is important in medical diagnostics, and how this is performed by using stethoscopes and by tapping on parts of the body. Several uses of sound and ultrasound have been described above as they relate to earlier chapters (e.g. listening to sounds, ultrasonography, echocardiography, ballistocardiography and so on), even though they are presented in this chapter. Ultrasound is also discussed in Problems 10.20–10.24. In addition, Chap. 10 presents the results of hearing sensitivity tests (Figs. 10.29, 10.42–10.45, 10.47 and 10.48) and ways of testing hearing (Fig. 10.46) and the immittance of the ear.

Chapter 11 describes testing visual acuity (Fig. 11.37) and several ways to treat imperfect vision by wearing corrective lensing and performing laser cornea sculpturing (Fig. 11.44). The curvature of the outer cornea surface can be measured with a keratometer (Fig. 11.42, Problem 11.73) and the intraocular pressure with a Goldmann tonometer (text and Problem 11.53); a goniolens can be used to inspect the eye (Fig. 11.61, Problem 11.12).

Measuring electrical signals in the body, a very important and widely used diagnostic, is detailed in Chap. 12. These signals are overviewed in Table 12.1. Perhaps of most importance are the electrical signals from the heart (electrocardiograms (EKGs), as in Figs. 12.28–12.38, Table 12.6), brain (electroencephalograms (EEGs), as in Fig. 12.39), and muscles (electromyograms (EMGs), as in Figs. 3.13, 3.31, and 12.22). Measuring magnetic fields can also be important (Table 12.8).

References

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